
EXERGYNET

Infrastructure for Useful Machine Work

Model-independent state and measured efficiency for AI systems that must persist across models, sessions, and organizations.

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Abstract

Artificial intelligence is becoming more capable and more substitutable at the model layer at the same time: new models arrive quarterly, model choice is increasingly a commodity decision, and the cost of a single inference call continues to fall. What has not fallen is the cost of everything a persistent, multi-step, multi-agent system must do around that inference call — reconstructing state, re-fetching evidence, coordinating multiple models and tools, and producing a record of what actually happened. That cost is paid in compute, memory, storage, network, energy, and engineering effort, and it does not disappear just because the model got better.

That gap motivates ExergyNet's first and most concrete question, which is economic before it is architectural: **can a system reduce the resource cost required to complete a validated useful task, measured against a workload's own existing, optimized baseline — without requiring that workload to abandon its chosen model or infrastructure?** On a single NVIDIA H200 with a Nemotron-class model, ExergyNet's xLMP memory layer held staged prompt cost roughly flat at ~660–820 tokens as the underlying corpus grew from 8,000 to 285,000 tokens, delivered a 24.4-point accuracy advantage over a tested RAG baseline at equal evidence budget, and reached roughly 11.3× the correct-task throughput of full-context replay at each method's best compliant load. These are bounded, tested-envelope results, not a universal savings claim, and they are reported that way throughout this paper.

Efficiency alone is not the product, and this paper does not present it as one. Reducing what a model must be shown for a given task is a mechanism — one that provider-native prompt caching, retrieval-augmented generation, and other memory systems are actively commoditizing. The durable requirement underneath it is different in kind: **models can change. Institutional state must persist.** An organization's authoritative record of what happened, on whose evidence, under whose permission, and with what provenance cannot be re-derived from a better model; it has to be built, maintained, and verified as its own thing, independent of which model is doing the reasoning this quarter.

ExergyNet is the architecture built to hold that requirement: a model-independent substrate that separates reasoning from authority, memory from authorization, and economic allocation from consequence control, organized into three coordination planes covering persistent state, coordination and authority, and resource accounting. This paper reports that architecture, its measured results within their tested envelopes, and the research frontier that remains open — including where execution-state portability across processes and machines is not yet solved, and where policy enforcement runs today in a monitored pilot rather than as a system's sole authority.

As machine intelligence becomes economically productive at industry scale — a scale now measured in tens of billions of dollars of quarterly infrastructure investment by the industry's largest supplier — the infrastructure connecting independent, heterogeneous intelligences to shared state, evidence, and consequential action becomes as consequential as the models themselves.

Executive Summary

The economic starting point. Model inference is getting cheaper and more interchangeable. The resource cost of persistent, state-heavy, multi-agent work is not falling at the same rate, because that cost lives in reconstruction, re-fetching, coordination, and verification — not in any single model call. This is the first question a buyer should ask about ExergyNet: does it reduce that cost, measured against the workload's own baseline, without requiring a change of model or cloud?

What ExergyNet does not require. See the box below — the short answer is: no required model change, no required cloud migration, no required exclusive vendor, no transfer of policy authority to ExergyNet, and no consequential permission that ExergyNet originates on its own.

The architecture. Once the economic case is established, ExergyNet's deeper product becomes visible: customer-controlled, model-independent authoritative state. ExergyNet sits at the boundary between a reasoning engine and the digital or physical systems it can affect (Figure 1), organized into three coordination planes (Figure 2): Knowledge & State, Coordination & Authority, and Resource & Economic Coordination. As models and agents multiply, that same state becomes the basis for interoperability across organizational boundaries; where actions become consequential, external authority and provenance become necessary. The organizing principle throughout is separation: reasoning and authority are never the same mechanism.

Four headline results, in the order a buyer should weigh them

Result	Measured outcome	Tested envelope
Useful-work / state efficiency	Prompt cost held ~660-820 tokens flat as corpus grew 8k-285k; full-context grew to 67k and was rejected past 262k	H200, synthetic corpus
State governance	False authoritative commitments fell 8% to 0%; model accuracy held at 84%	100 real procurement cases
Deterministic authority	Rejected a live prompt-injection after the reasoning model complied; reproduced twice	Live listener, unit-tested
Physical consequence	False-release held at 0/50 in two independent runs vs. 20-34% ungoverned	Sealed synthetic holdout

The larger implication. As machine output becomes economically productive at industry scale, the resources and coordination mechanisms that sustain it become strategic infrastructure in their own right, not implementation detail.

What ExergyNet Does Not Require

ExergyNet does not inherently require an adopting organization to:

- replace its frontier model or reasoning engine of choice;
- adopt one exclusive model vendor;
- move all enterprise data into an ExergyNet-owned store;
- transfer policy authority to ExergyNet; or
- allow ExergyNet to originate consequential permissions on its own.

Exception, disclosed: opt-in settlement through ExergyNet's own test-network infrastructure (Section 9) does involve an ExergyNet-operated component; it is not a requirement of adopting the memory, state, or authority layers described elsewhere in this paper.

Each point above is a property of the architecture described in this paper, not a marketing assertion: Section 3 shows the model as an interchangeable component at the boundary, not a replaced one; Section 4 introduces the Context/Memory/Authoritative-State distinction that keeps a customer's own data as the thing being preserved, not relocated; Section 6 states plainly that authority must originate externally from the institution using the protocol.

1. The Economic Problem

Every independent reasoning system begins each session from nothing durable: state is reconstructed at full cost, evidence is re-fetched without stable identity, authority is implicit in whatever credentials the caller holds, and no independent record survives to explain why an action occurred. This is a property of the surrounding architecture, not of any particular model, and improving the model does not change it. The cost is not paid once — it is paid every session, every model swap, and every hand-off between agents, in compute, memory, storage, network, energy, and engineering time.

This argument is no longer only architectural — it is now visible in the infrastructure economics of the companies building frontier compute. In its most recent quarter, NVIDIA reported \$89.0 billion in Data Center revenue, up 117% year-over-year, and supply and capacity purchase commitments that rose from \$119 billion to \$279 billion quarter-over-quarter — an increase its finance leadership attributed primarily to the procurement of memory. At the same time, independent technical disclosures from memory manufacturer Micron reported that compute performance is scaling roughly 3× every two years while high-bandwidth memory (HBM) capacity is scaling less than 2× over the same period (Figure 9).

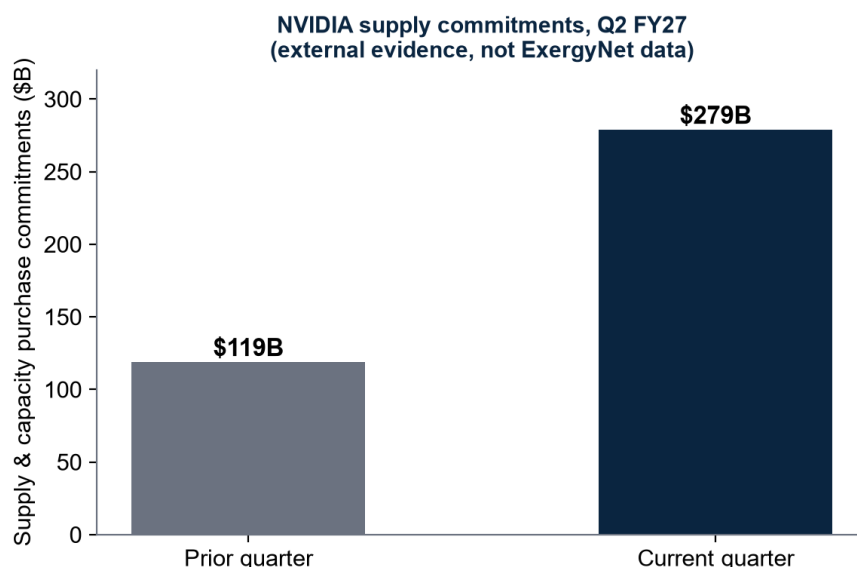


Figure 9a. NVIDIA supply and capacity purchase commitments, Q2 FY2027 (external evidence, not ExergyNet data).

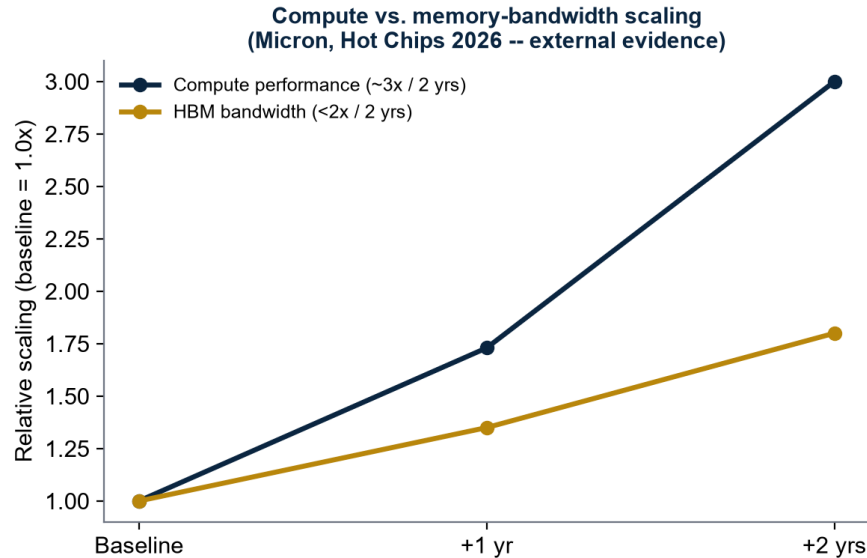


Figure 9b. Compute scaling vs. HBM capacity scaling, Hot Chips 2026 (external evidence, not ExergyNet data).

These figures are external, dated evidence about the industry, not ExergyNet results, and they do not validate this architecture. What they confirm is the premise this paper derives independently: memory, compute, energy, and physical infrastructure have become contended, strategically managed resources — exactly the resource layer that a workload's own cost-per-validated-task must eventually measure, allocate, and account for (Sections 2 and 9).

The question this paper answers first is therefore economic, not architectural: given a workload an organization already runs, on infrastructure it already operates, can the resource cost of completing a validated useful task go down — measured against that workload's own existing, optimized baseline — without requiring a different model or a different cloud? Section 2 makes that question precise. Section 4 reports the measured result. Only after that does this paper introduce the architecture that makes the result durable rather than a one-time optimization.

2. Useful Work: A Measurement Layer for Machine Intelligence

Before describing the architecture, this paper defines the commercial measurement problem it is built to answer. Raw token count is not the right final denominator — a workload that halves its tokens but doubles its retries or triples its coordination overhead has not become more efficient. The correct denominator is the resource cost of a **qualified successful task**: one whose output meets whatever correctness or validation bar the workload itself already requires.

Define the cost per qualified successful task as:

$$C_Q = \frac{C_{\text{total}}}{N_{\text{qualified successful tasks}}}$$

where total cost includes, where measurable:

$$C_{\text{total}} = C_{\text{inference}} + C_{\text{retrieval}} + C_{\text{state}} + C_{\text{verification}} + C_{\text{retries}} + C_{\text{coordination}} + C_{\text{network}} + C_{\text{storage}}$$

and, symmetrically, define useful-work efficiency as the inverse ratio against total energy consumed:

$$E_Q = \frac{N_{\text{qualified successful tasks}}}{E_{\text{total}}}$$

alongside the completion-time distributions T_{p50} and T_{p95} , since a workload's tail latency is often as commercially relevant as its mean cost.

Currently measured

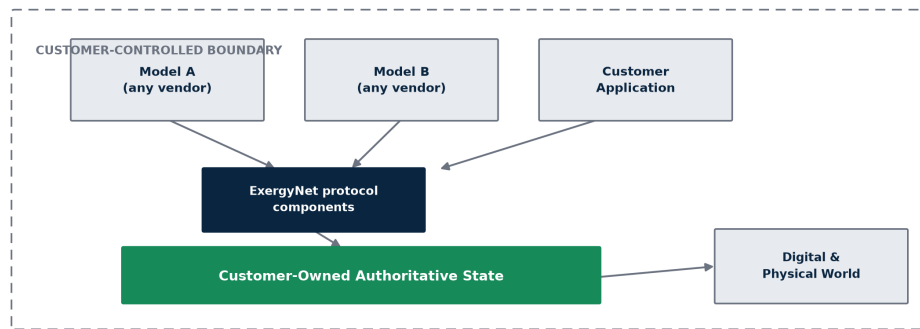
The results reported in Section 4 measure specific terms of C_{total} directly — principally $C_{\text{inference}}$ and C_{state} , via staged prompt-token count as a cost proxy on a fixed hardware target (a single NVIDIA H200). They do not yet measure C_{retries} , $C_{\text{coordination}}$, C_{network} , or C_{storage} as independent, itemized terms, and E_Q is not separately measured in this paper — energy is not directly instrumented in the reported results.

Proposed standardized evaluation framework

C_Q , E_Q , and the full multi-term cost decomposition above are presented here as the target evaluation framework for the benchmark program this paper's research frontier describes (Section 12), not as an existing, fully-instrumented result. ExergyNet has not measured every term in C_{total} for any reported workload, and this paper does not claim otherwise. The framework itself is architectural and research framing: a standard against which future, more complete measurements can be reported and compared, workload by workload, against each workload's own optimized baseline — never as a universal efficiency multiplier.

3. The ExergyNet Architecture

ExergyNet occupies the boundary between a reasoning engine and the world it can affect. The reasoning engine is interchangeable — GPT, Claude, Nemotron, an open-weight model, a robotics controller, or a human operator — and the architecture is drawn accordingly: models and applications connect through ExergyNet's protocol components to state the customer owns, inside a boundary the customer controls (Figure 1). ExergyNet does not hold that state on the customer's behalf as a precondition of using it, and it is not drawn as a box every request must pass through with no visibility into what happens inside.



Models connect through ExergyNet protocol components to state the customer owns and controls. ExergyNet does not hold the state; it is not a required intermediary the customer cannot see or replace.

Figure 1. Models connect through ExergyNet protocol components to state the customer owns and controls.

The architecture organizes into three coordination planes (Figure 2):

- **Plane I — Knowledge & State.** What exists, what happened, and what is currently authoritative: xLMP / Exergy Vault, deterministic indexing and routing, Temporal Authority, and physical observation (Edge Witness).
- **Plane II — Coordination & Authority.** Who may interact with what, under what constraints: machine identity, the xISA capability vocabulary, Vanguard model routing, the LNES-22 Consequence Boundary, and AERIS external-evidence acquisition.
- **Plane III — Resource & Economic Coordination.** What was consumed, authorized, and settled: RHO resource accounting, Omega, and MMS settlement.



Plane membership is an organizing view, not a claim that every listed component is deployed -- see Section 9.

Figure 2. The three coordination planes.

The architecture keeps a small set of predicates permanently separate: memory evidence is not execution authority; model identity is not agent identity; agent identity is not economic authority; economic authority is not consequence authority; and resource accounting is not market value. These separations are what let intelligence be swapped, fail, or be distrusted without the surrounding system losing what it knows, what it may do, or what it must account for.

Security posture. ExergyNet assumes the model may be compromised. Consequential authorization is placed outside the reasoning model so that protected actions can be evaluated deterministically, independent of whether the model that proposed them can be trusted (Section 8).

4. Persistent State and xLMP

The clearest cost of statelessness is reconstruction: an agent that rebuilds its world every time it changes model, machine, or session is, in effect, rebuilding the road behind itself every time it moves. xLMP is ExergyNet's answer — a root-addressed, content-verified persistent evidence store with bounded retrieval, and the paper's primary measured answer to the economic question posed in Sections 1 and 2.

On a single NVIDIA H200 with a Nemotron-class model over a synthetic corpus, xLMP held staged prompt cost roughly flat at ~660-820 tokens as the underlying corpus grew from 8,000 to 285,000 tokens, while full-context injection grew to 67,000 tokens and was rejected outright past 262,000 (Figure 4). At equal evidence budget, xLMP retrieval reached 92.4% accuracy against a 68.0% tested RAG baseline — a 24.4-point advantage — and delivered roughly 11.3× the correct-task throughput of full-context at each method's best compliant load (Figure 5).

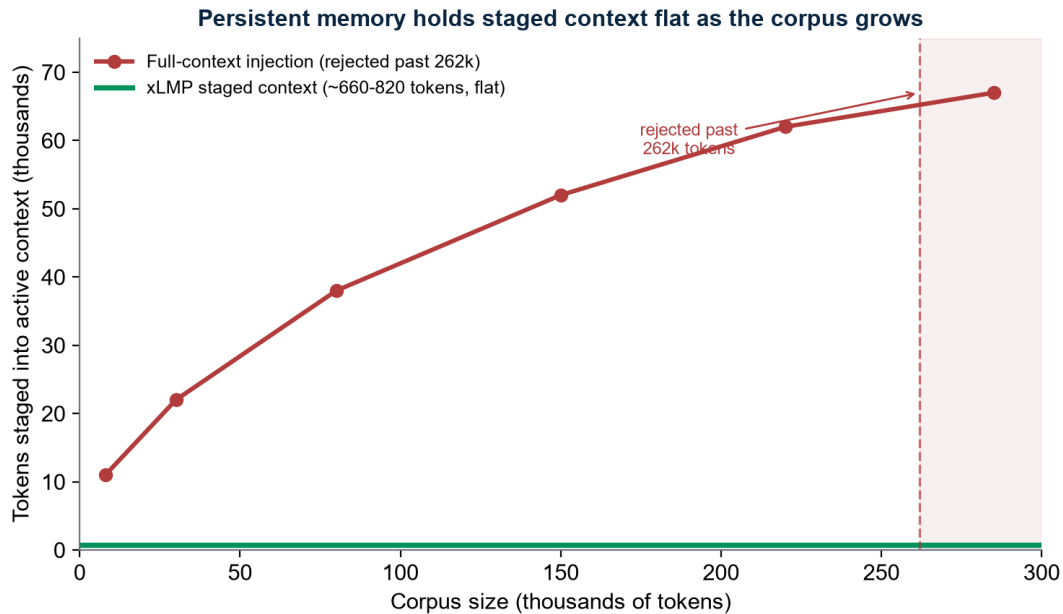


Figure 4. Staged prompt-token cost vs. corpus size, xLMP vs. full-context replay.

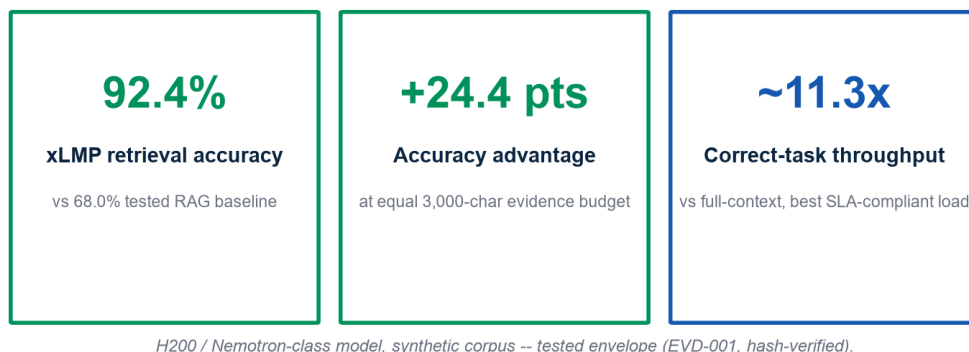


Figure 5. Accuracy and correct-task throughput, xLMP vs. tested RAG baseline and full-context replay.

Why this is a capacity result, not only a cost result. Reducing the context staged into active working memory per operation means the same physical hardware sustains more useful machine work per unit of memory and time. Under constrained memory and compute, that is a capacity property, not merely a savings — and it is precisely the C_{state} term of the useful-work cost decomposition introduced in Section 2.

Validation scope. These results hold within the tested H200 / synthetic-corpus envelope; broader-scale validation is part of the ongoing measurement program described in Section 12. A separate efficiency figure (~42.7x vs. full-context, ~4.0x vs. RAG) is reported pending independent re-verification of its source artifact.

Context, Memory, and Authoritative State

Efficiency mechanisms — caching, compression, retrieval — reduce what a model is shown for a given task. They do not, by themselves, answer a different question: what is the institution's current, canonical record of what is true, and who is allowed to change it? This paper draws a firm line between three concepts that are commonly blurred:

- **Context.** Information supplied to the reasoning engine for the current inference. Exists only for the duration of that call.
- **Memory.** Information retained because it may be useful in later reasoning. Improves recall; does not, by itself, establish what is canonical.
- **Authoritative State.** The externally maintained representation of what the system currently treats as canonical, including, where applicable: version, source, provenance, authority, evidence, and transition history.

Context $\neq$ Memory $\neq$ Authoritative State

This distinction is commercially load-bearing, not academic: provider-native prompt caching and memory systems are rapidly commoditizing the first two categories. A vendor that reduces token cost or improves recall has not, by that fact alone, produced a system that knows what is currently true, who said so, and under what permission it may be acted on. This paper does not make claims here about any specific competing system's architecture; the distinction is stated on its own terms, and the reader can apply it to whatever alternative is under evaluation.

5. State Mobility

An autonomous agent's active execution state — not just its stored memory — is itself a resource worth preserving. ExergyNet treats execution state as a first-class object with its own validation envelope (Figure 10).

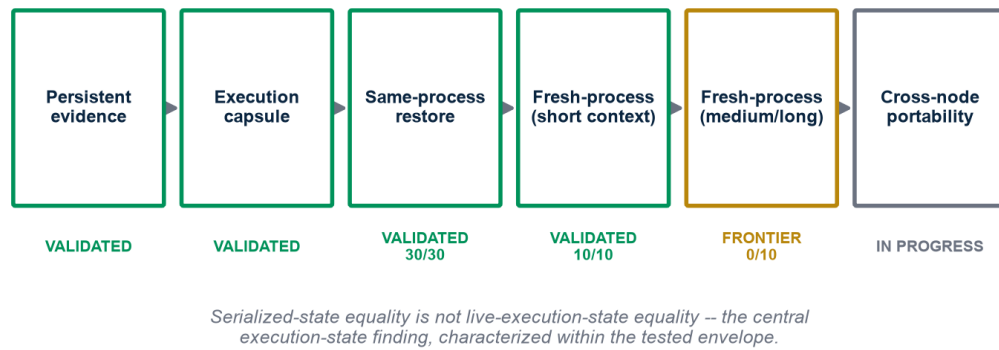


Figure 10. Execution-state validation envelope: same-process, fresh-process, and cross-node continuity.

On a hybrid KV/SSM execution runtime, same-process erase-and-restore reproduced continuation exactly across every tested context length (30 of 30 aggregate trials), with bit-perfect capsule round-trips (9 of 9 checked). True fresh-process restoration reproduced continuation exactly for short contexts (10 of 10) — the validated frontier today. For medium and longer contexts, fresh-process continuation diverged in the tested runs, isolating a genuine research finding: **serialized-state equality is not the same as live-execution-state equality**. The capsule bytes themselves were bit-perfect; the divergence has been localized to runtime cache-reconstruction semantics, which remains under active investigation.

Architectural objective vs. current measured portability

	Statement
Architectural objective	Portable, externally authoritative, verifiable state across model and administrative boundaries.
Current measured execution-state portability	Same-process restoration: validated. Fresh-process, short-context: validated. Fresh-process, medium/long-context: not yet achieved in tested runs (0/10 each). Cross-node: not yet reported.

The objective is the reason this architecture exists. The measured column is what has actually been shown so far. Neither is presented as the other.

Validation scope. Same-process continuity and short-context fresh-process continuity are validated. Medium/long-context fresh-process continuity and cross-node portability are the active research frontier — no cross-node success is reported until an in-progress validation campaign is complete.

6. The Model-Substitution Principle

A model-independent substrate is only as valuable as its answer to a specific, concrete event: a model is replaced. Not hypothetically — routinely, as pricing, capability, and vendor relationships change.

$$M_A \rightarrow M_B \rightarrow M_C$$

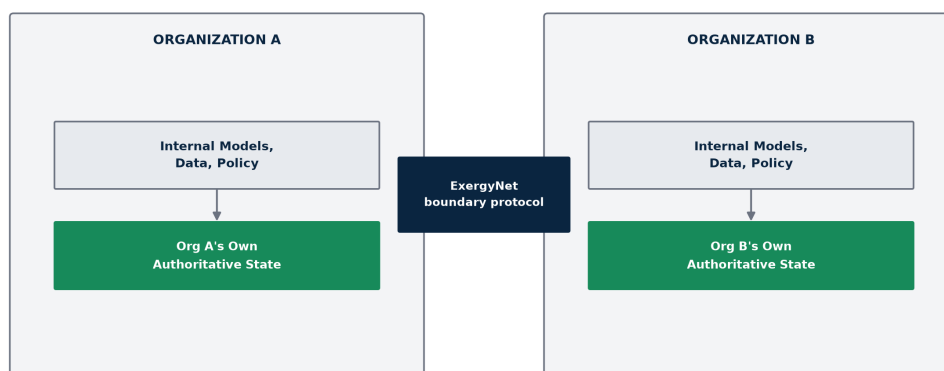
while $S_{\text{institution}}$ must remain continuous.

Models can become interchangeable without institutional truth becoming disposable.

This is an architectural requirement this paper states plainly, not a result this paper claims to have already fully achieved: Section 5 already reports that full runtime-state portability across processes and machines is not yet solved for every tested envelope. What is architecturally in place is the separation itself — the authoritative state described in Section 4's boxed distinction is defined independently of any one model's internal representation, which is what makes the substitution principle a coherent target rather than a slogan. The point of this section is the requirement the architecture is built to satisfy, not a claim that the hardest form of it — silent, lossless substitution under any workload — has already been demonstrated end-to-end.

7. Administrative Boundaries and Interoperability

Some of the strongest requirements this architecture answers only appear once more than one administrator is involved. Consider a scenario with no single trusted administrator: a company evaluating a vendor's model output, a government agency requiring an audit trail independent of the vendor, a model provider that cannot see the customer's internal data, and a cloud provider that hosts infrastructure for both without adjudicating disputes between them. None of these parties can be asked to simply trust another's internal state.



Each organization retains its own internal systems and authoritative state. ExergyNet coordinates only the boundary interaction between them — it is not inside either organization.

Figure 3B. Two organizations retain their own internal systems; ExergyNet coordinates only the boundary interaction.

Each organization retains its own internal systems and its own authoritative state, bridged only at the interaction boundary — not merged, not placed under one administrator, and not required to expose their

internal systems to each other or to ExergyNet beyond what the interaction itself requires. This is the stronger version of a “neutral ground” position: not an absolute claim about corporate neutrality, but a property demonstrated through the architecture itself — verifiable state and evidence exchanged at the boundary, nothing more.

The integration-cost argument, correctly bounded

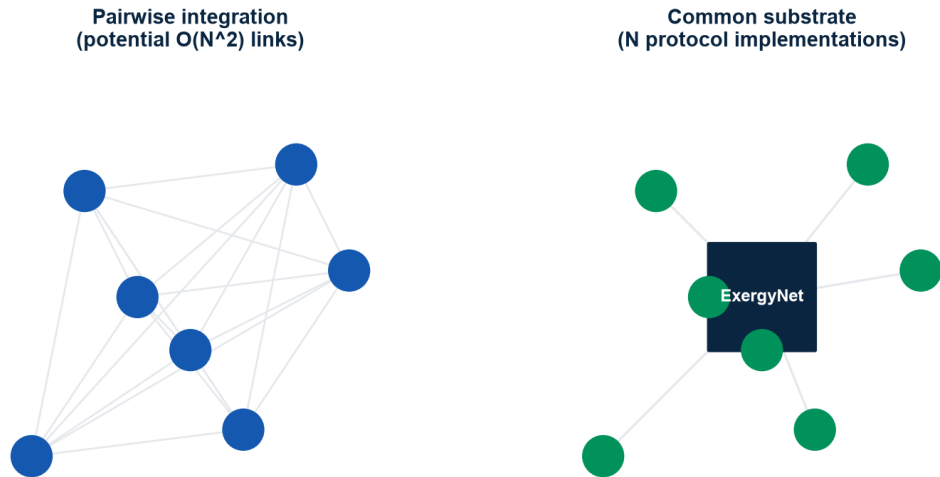


Figure 3A. Pairwise integration (potential $O(N^2)$ links) vs. a common substrate (N protocol implementations).

As the number of independent, heterogeneous intelligences and organizations (N) in a shared workflow grows, unconstrained pairwise integration between every pair carries a potential quadratic cost; a shared substrate instead lets each participant implement one common interface. This is a potential integration bound, not a measurement of actual network traffic in any deployed system, and it does not imply that every model needs ExergyNet. The defensible conclusion is narrower and remains true even in a world where most participants never adopt this protocol:

If independently governed systems require repeated bilateral state, identity, evidence, or authority integrations, common protocol semantics can reduce the bespoke integration burden those systems would otherwise each pay separately.

8. Reasoning, Evidence, and Authority

Reasoning and authorization are architecturally separate. A proposed action must independently clear four gates — identity, capability, freshness, and integrity — before it is treated as authorized; failing any one gate is a denial (Figure 7).

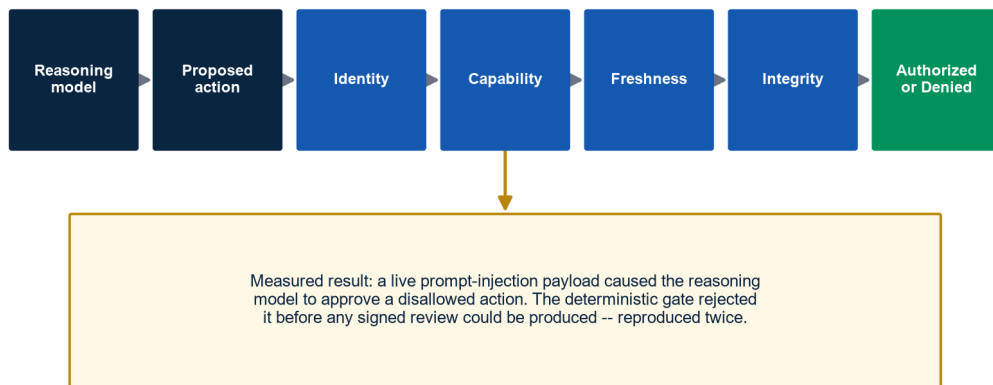


Figure 7. The four-gate authorization pipeline.

The strongest result. ExergyNet sent a live prompt-injection payload to a running authority-review listener. The reasoning model partially complied: it returned an approval and did not flag the injection despite its own instructions to do so. The deterministic validation layer rejected the request outright before any signed review could be produced — no credential was exposed at any point, because key material is structurally excluded from what reaches the model. The attack was reproduced after a logging fix, with the same rejection now producing a durable, signed record. The finding, bounded to this validated run and stated as evidence for one specific architectural principle — **reasoning and authority must remain separable, not that ExergyNet should govern every model** — is this: a reasoning model can fail adversarially without the authority system failing with it.

Supporting validation: 64 assertions across three independent unit-test scripts; 38 of 39 adversarial tests passed against the agent-authorization API (the one failure was a test-environment artifact); and 100 of 100 concurrent double-allocation attempts against a shared balance were correctly serialized.

Evidence integrity is separately governed. State-governance testing over 100 real procurement-decision cases reduced false authoritative-state commitments from 8% to 0% while model judgment accuracy held flat at 84% (Figure 6) — the substrate prevented unresolved evidence from being promoted into an authoritative record, within the tested case set. This distinction matters commercially: **the model did not become more accurate; the surrounding system prevented unresolved evidence from becoming authoritative state.** Better governance is not the same thing as better model intelligence, and this result is evidence for the state-centric product definition in Section 4, not for the model's own reasoning quality.

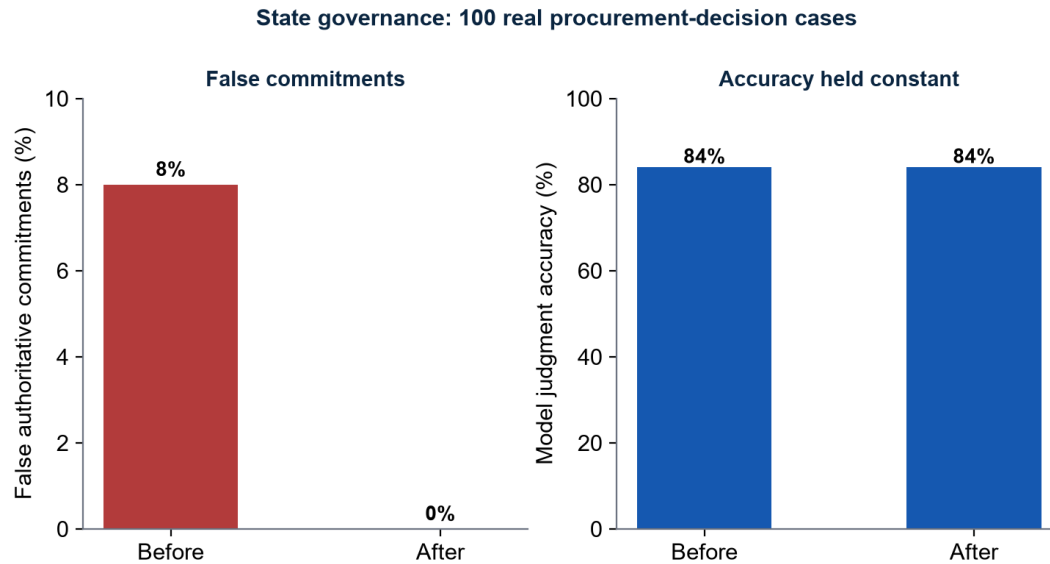
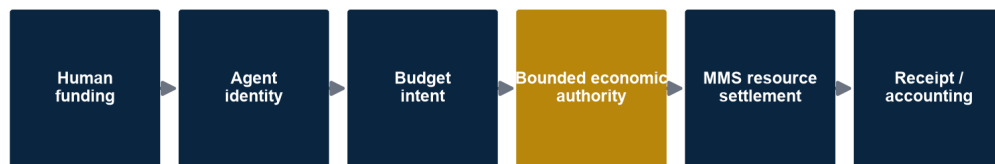


Figure 6. False authoritative-commitment rate, governed vs. ungoverned, 100 procurement cases.

Validation scope. The identity/capability/freshness/integrity validator is live and independently tested. The broader policy-capability gate is implemented and tested but currently runs in a monitored pilot mode: it evaluates and logs every real decision without yet being the sole authority over execution (Section 12).

9. Machine Resource Economics

If machine output is economically productive, the resources consumed to produce it — compute, memory, storage, network, energy, and time — become economically significant in their own right. ExergyNet's answer is a resource-authority pipeline: humans fund an agent identity, which receives bounded economic authority, spends it under explicit policy, and produces an auditable settlement record (Figure 11).



Demonstrated end-to-end on Base Sepolia testnet: 500 RHO settled, every replay reverted.

Figure 11. The Omega resource-authority pipeline: fund, authorize, spend, settle.

RHO is ExergyNet's resource-cost accounting unit, defined by measurement, not by market price. A metrology pass fixed a measured cost basis across two hardware classes; the operations with the lowest

cost are measured directly at the host, while the highest-cost operation (dominated by GPU inference) is currently modeled from pricing data rather than measured on-device — the next stage of the metrology program. RHO's value is defined independently of any market price, token, or energy interpretation.

RHO is not the same denominator as useful-work efficiency. Section 2's C_Q measures the resource cost of a validated task against a workload's own baseline; RHO is a separate, internal resource-accounting semantic used for settlement between funded agent identities. A reader should not infer cryptocurrency-style token economics from this paper's commercial thesis: RHO has no defined market price, is not redeemable for a fixed unit of currency, and is not presented anywhere in this paper as evidence for or against the useful-work efficiency argument made in Sections 2 and 4.

Demonstrated settlement. A full three-operation economic strike — recall, write, and query — settled end-to-end on a public test network: per-model spending allowances were granted, spent to zero, and every replay attempt was reverted, with 500 RHO settled and independently recomputed from two different accounting paths.

Validation scope. Settlement is demonstrated on testnet; production-scale, multi-principal settlement and any market valuation of settled units are the next stage.

10. When Software Becomes Structural

As machine intelligence moves from producing information to controlling consequential systems — a financial transfer, an infrastructure change, a physical action — the software governing that transition stops being an implementation detail and becomes part of the system's load-bearing structure, in the way a bridge's structural members, not its paint, keep it standing. A structure that is correct on average is not sufficient when a specific, consequential transition is underway: the governing invariants must hold for *that* transition. This is the architectural reason reasoning and authority cannot be the same mechanism (Section 8).

The primary consequence-active example: an aviation pre-flight authorization gate. A heavy-lift uncrewed aircraft program illustrates the requirement with unusual clarity: a physical machine with real authority to release cargo or return to service, continuous telemetry, explicit safety invariants, and consequences that cannot be undone once actuated. ExergyNet's authorization gate for this program produces exactly three terminal states — release-eligible, hold, or incomplete — as an additional safety layer, deferring final authority to its own separate policy evaluation (Figure 12).



Sealed 50-case synthetic holdout: governed false-release held at 0/50 across two independent runs, vs. 20-34% for ungoverned telemetry. Simulated witness data only.

Figure 12. Aviation pre-flight authorization gate: telemetry to terminal state.

Across two independent validation runs — a deterministic rule-based simulator and, separately, a real reasoning model — the governed gate held false-release at 0 of 50 cases on a sealed synthetic test set, against a 20-34% false-release rate for the same cases evaluated on ungoverned raw telemetry (Figure 8).

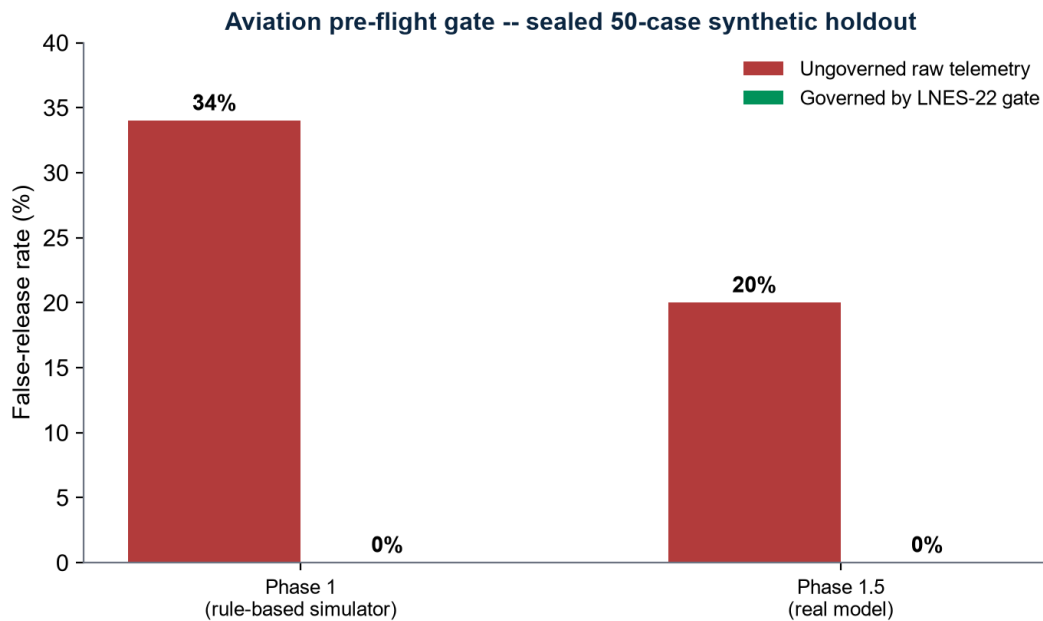


Figure 8. False-release rate: governed gate vs. ungoverned raw telemetry, sealed 50-case holdout.

Validation scope. This result is measured against a sealed, synthetic 50-case holdout; no real aircraft or sensor hardware was involved. A regulatory authorization exists for the related aircraft program permitting controlled testing and evaluation — a fact about that program's regulatory status, not a certification of this architecture.

11. Applications

The architecture stands independently of any one vertical; removing this section leaves it intact.

Aviation (Section 10) is the primary demonstrated consequence-active application. **Clinical decision support** is a second, illustrative case study: a clinical assistant must preserve an authoritative patient-state record across sessions, must not promote unresolved evidence into that record, and must keep model reasoning separate from any consequential action — properties this architecture is built to provide, presented here as an illustrative reference pattern rather than a deployed clinical integration.

Robotics and critical infrastructure, scientific and research agents, industrial automation, enterprise autonomous agents, financial and transactional agents, and secure machine-to-machine communications share the same underlying requirement — an identity-bound, evidence-backed, capability-scoped decision before a consequential action — and are architecturally applicable under the same protocol, without a bounded internal benchmark reported for each.

12. Validation Status and Research Frontier

Every component below is reported at its actual, current maturity stage: **Defined** (specified) → **Implemented** (built) → **Validated** (reproducibly exercised in a test envelope) → **Pilot / Shadow** (running against real decisions, not yet the sole authority) → **Production** (operating as the system of record).

Component	Maturity	Boundary
xLMP / Exergy Vault	Production	Published local node (VMN); integrity is distinct from correctness
Compact Index / Adaptive Routing	Validated → Partial production	Performance is workload-dependent
Vanguard (model routing)	Production	Proposes actions; does not self-authorize
Authority validator	Production	Live-validated against a real adversarial run
Policy / capability gate	Pilot	Evaluates and logs every real decision; not yet sole execution authority
xISA capability taxonomy	Validated	Isolated research environment; not yet production-active
Temporal Authority	Implemented, tested	Not yet production-activated
RHO resource metrology	Pilot	Not monetary; GPU-cost term currently modeled, not measured
Omega	Implemented	No live spending authority currently granted
MMS settlement	Validated	Test network only
AERIS (external evidence)	Validated	Test network; production settlement requires a further generation

Edge Witness (physical observation)	Production	Attests device signature, not physical truth
Execution-state continuity	Validated (same-process, short fresh-process)	Medium/long fresh-process and cross-node are the active frontier
Aviation pre-flight gate	Validated (synthetic holdout)	No real-aircraft validation yet reported
Physical-state observation (forward architecture)	Defined (proposed)	No implementation yet

Research frontier, stated plainly: production-wide authority enforcement across every live route; true cross-process and cross-node execution-state portability; a physical-state observation primitive; and production-scale zero-knowledge verification integrated into the evidence-query path. These are named as frontiers because they are not yet true, not because the architecture cannot eventually reach them.

Proposed Comparative Evaluation Program

The useful-work measurement layer introduced in Section 2 motivates a standing comparative benchmark program against a common set of reference baselines:

- B_0 — full-context replay
- B_1 — provider-native caching
- B_2 — optimized RAG
- B_3 — persistent/provider memory
- B_4 — ExergyNet / xLMP
- B_5 — native optimization + ExergyNet

Section 4 reports results consistent with B_0 and B_2 (full-context replay and a tested RAG baseline) as T1, tested-envelope results. **No result is reported for B_1 , B_3 , or B_5 until an artifact exists in the claim ledger.** This program matters commercially because the relevant comparison for an adopting organization is not xLMP against an intentionally expensive replay architecture — it is xLMP's incremental value after that organization's own already-optimized baseline, which is exactly what B_5 is designed to measure once run.

Conclusion

The AI industry is building the vehicles: increasingly capable, increasingly numerous, increasingly heterogeneous models, built by frontier labs, open-model communities, and specialized vendors alike. That buildout is now measured in tens of billions of dollars of quarterly infrastructure investment and, by its largest supplier's own account, still accelerating. The models do not need to become interchangeable for that buildout to matter here — model plurality is precisely what makes shared infrastructure necessary, not optional.

**The AI industry is building the vehicles.
ExergyNet is building the roads required when those vehicles become
numerous, heterogeneous, persistent, and autonomous.**

More precisely: ExergyNet's objective is a common machine infrastructure in which intelligence can persist, establish identity, preserve evidence, exercise bounded authority, account for resources, and interact with the physical world — without requiring those functions to reside inside any single model. Its contribution is a substrate that keeps knowledge, evidence, capability, authority, economy, and execution state as separate, independently verifiable properties, so that intelligence can be swapped, can fail, or can be distrusted without the surrounding system losing what it knows, what it may do, or what it must account for. If every current frontier model were replaced tomorrow, the reason ExergyNet exists would be unchanged.

Design partners can provide a bounded, state-heavy workload for comparative measurement against their existing optimized baseline. **Benchmark the workload, not believe the claim.**

References

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6. U.S. Provisional Patent Application No. 64/134,973, filed 2026 (formal claims subject to prosecution).
7. Federal Aviation Administration. Exemption No. 26214, Docket FAA-2025-5731.

A full claim ledger, source registry, and revision history are maintained as separate internal diligence documents and are available on request.